Know It's Absent, Yet Point Anyway

Cross-Task Semantic Contradictions in Vision-Language Models

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Introduction & Motivation

Rapid Rise of Multimodal NLP

Vision-Language Models (VLMs) have revolutionized multimodal NLP, combining:

- Generative NLP reasoning
- Visual Question Answering (VQA)
- Spatial Grounding (connecting text tokens to pixel coordinates/bounding boxes)

The Semantic Consistency Challenge

Despite high VQA performance, VLMs exhibit severe **cross-task logical inconsistencies**:

- They correctly recognize object absence in text format...
- ...yet fail to preserve this semantic judgment when the task is reformulated as a grounding prompt.

Key Question: How can a model that correctly answers "No" to "Is there a cat?" still draw a bounding box when later asked to "Point to the cat"?

Part 1: The Phenomenon – Existence-Localization Contradiction

The Core Contradiction

A systematic failure mode where a VLM:

1. Correctly recognizes that a queried object is **absent** from an image,
2. But still produces a **non-null, hallucinated localization** (bounding box) when asked to ground the same referent.

This isolates cases where models know the referent is absent but fail to carry this judgment into grounding.

Actionable Hallucination in NLP Agents

A physical grounding commitment contradicts the internal semantic belief.

- Critical for **downstream NLP systems** (grounding-based tool use, database retrieval, multimodal web agents).
- Propagates false evidence, triggering incorrect downstream actions in multimodal pipelines.

Part 1: A Concrete Example (Figure 1)

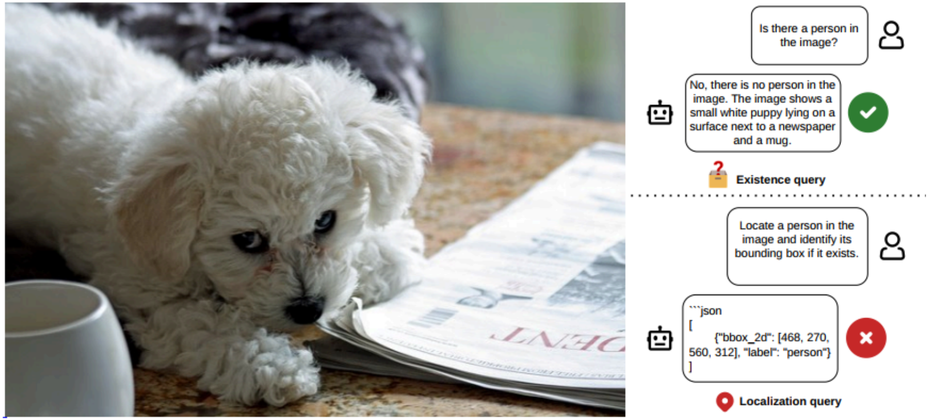

Figure 1: An example of Existence–Localization Contradiction.

Figure 1: VLM correctly recognizes object absence in VQA but localizes it anyway when prompted.

2-Stage Conditional Protocol

To isolate contradictions, the authors evaluate localization **only** when the model correctly identifies absence:

Stage 1: Existence Recognition

- Query VLM with existence prompt $T_E(o_i)$ on negative set (absent objects).
- Filter cases with correct absence ($\hat{e}_i = 0$).

Stage 2: Localization under Absence

- Query the VLM with localization prompt $T_L(o_i)$.
- Assess if it returns a null output ($\hat{\ell}_i = 0$).

Primary Metric: CELF

Conditional Existence-Localization Faithfulness (CELF):

$$\text{CELF} = \Pr(\hat{\ell}_i = 0 \mid e_i = 0, \hat{e}_i = 0)$$

CELF measures how often the model remains faithful to its prior correct absence judgment during localization.

- **CELF = 1.0:** Perfect consistency.
- **CELF = 0.0:** Complete failure (always localizes anyway).

Part 2: Prevalence of Contradictions in Strong VLMs

Table: Baseline CELF results on POPE, AMBER, MME, DASH-B

Model	Metric	POPE	AMB	MME	DASH
Qwen2.5-7B	EA ↑	83.7%	95.4%	100%	74.3%
	CELF ↑	**10.4%**	**28.6%**	**20.0%**	**2.8%**
Qwen3-8B	EA ↑	89.4%	90.2%	98.3%	74.5%
	CELF ↑	**15.8%**	**38.2%**	**50.0%**	**2.7%**
InternVL3	EA ↑	90.8%	92.3%	98.3%	67.6%
	CELF ↑	**0.0%**	**0.0%**	**0.0%**	**0.0%**

Key Observation

While Existence Accuracy (EA) is consistently high $(80\% - 100\%)$, **CELF scores are strikingly low**.

InternVL models achieve **0.0% CELF** across all datasets, meaning they **never abstain** under localization queries, even when they just correctly recognized object absence.

Part 2: Representation Probing – Why Do Contradictions Occur?

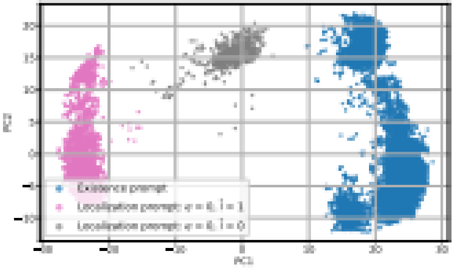

Figure 2: PCA projection of hidden states.

Figure 2: PCA projection of hidden states at the final token position.

Task Representation Separation

- Hidden states under **existence prompts** and **localization prompts** form two highly distinct clusters.
- Represents different internal “regimes” with very limited multi-task knowledge sharing.

Proximity of Correct Rejections

- Correct null-localizations lie significantly **closer** to the existence prompt cluster.
- Rejection succeeds when the VLM manages to leverage prior existence knowledge.

Part 3: Solution – Representation Steering (Activation Engineering)

The authors propose **inference-time activation steering** to shift hidden states towards absence-consistent representations (Parameter-Free, localized, and efficient):

Step-by-Step Methodology

1. **Paired Contrastive Set:** Build a set $\mathcal{P}$ of 100 examples. For each query, extract hidden states at transformer layer l for the hallucinated response $h_{loc}^{(l)}$ and the correct abstention response $h_{abs}^{(l)}$.
2. **Compute Steering Direction Vector ($s^{(l)}$):** Define the mean difference vector (similar to LLM truthfulness steering):

$$s^{(l)} = \frac{1}{M} \sum_{i=1}^{M} \left(h_{i,abs}^{(l)} - h_{i,loc}^{(l)} \right)$$

3. **Inference-Time Injection:** Add the steering vector into the hidden representation during test time at layer l:

$$\tilde{h}^{(l)} = h^{(l)} + \alpha s^{(l)}$$

Part 3: CELF Performance under Different Steering Scales

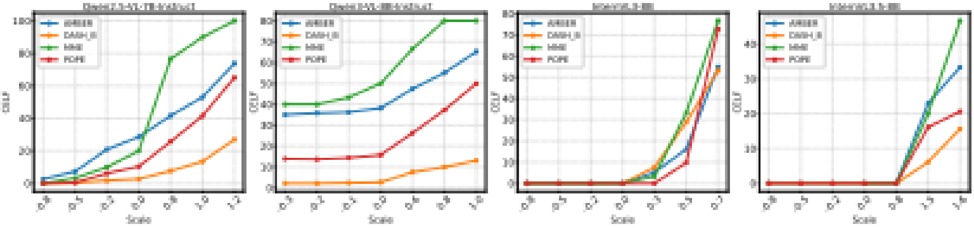

Figure 3: Model performance with different steering coefficients.

Figure 3: Model performance (CELF metric) increases steadily as positive steering scale α rises.

Conclusion & Outlook

Summary of Contributions

- **Identified Contradiction:** Uncovered Existence-Localization Contradiction as a systematic semantic logic failure in leading VLMs.
- **Conditional Protocol:** Formulated the CELF metric to evaluate consistency.
- **Activation Steering:** Demonstrated that lightweight activation steering dramatically restores absence-consistent grounding.

NLP Research Perspectives

- **Interpretable NLP:** Highlights the power of representation probing and steering to dynamically align VLM behaviors without fine-tuning.
- **Architectural Coherence:** Encourages designing multi-task pretraining or loss formulations that bind text reasoning and spatial grounding logic, rather than treating them as isolated tasks.